

Verdant Coil – Full Game Overview

Concept Summary

Verdant Coil is an **asynchronous, tile-based dungeon builder and explorer game** set in a surreal, biotech-inspired world. Players design and publish living maze-like dungeons called *Coils*, filled with hazardous, evolving organic structures. Others explore these Coils by guiding a customizable creature (a *Crawler*) through the maze to the *Heartroot*.

Player interactions are **asynchronous** and indirect—explorers earn primary resources by completing Coils, while creators earn secondary resources when their Coils are successfully explored. The game encourages ongoing engagement through creative expression and discovery, with a resource-based progression loop.

Core Gameplay Loop

1. Explore Mode (PvE)

2. Control a Crawler in a **turn-based**, top-down grid world.
3. Traverse living mazes built by other players.
4. Encounter and adapt to hazards, terrain, and puzzle-like obstacles.
5. Collect Nutrients while navigating the level.

6. Reach the *Heartroot* to complete the level and earn **Nutrient**.

7. Build Mode (Creator)

8. Design maze-like Coils on a 2D grid using modular biological tiles.
9. Use a **biomass budget** to control tile placement and enforce creativity under constraints.
10. All levels must be validated by the creator (completed successfully) before publishing.
11. Published levels are saved locally (initial MVP) or shared via backend (future).

12. Earn **Sporeprint** passively when other players complete your levels.

13. Progression

14. **Nutrient**: Earned by playing; used to unlock upgrades and expand tile-building options.
 15. **Sporeprint**: Earned passively when your levels are completed; used for advanced tiles, higher biomass caps, and cosmetic enhancements.
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Player Unit – The Crawler

- Controlled in discrete tile-by-tile turns.
- Movement is deterministic and based on player input.
- Base stats include:
 - **Health**: Decreases when exposed to hazards.

- **Armor:** Mitigates or nullifies specific damage types.
 - **Speed:** Affects delay between tile transitions (if implemented).
 - **Traction:** Counteracts slow effects.
 - **Perception:** Reveals hidden tiles and hazards.
 - **Acid Reserve:** Limits how many Digest Walls can be dissolved per run.
 - Toggleable Upgrades:
 - **Hardened Skin:** Reduces acid damage.
 - **Acid Sac:** Dissolves Digest Walls.
 - **Photoreceptors:** Detect hidden tiles and paths.
 - **Directional Memory:** Recall path history to aid navigation.
 - **Locomotion Shift:** Navigate sticky or goo terrain.
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✳️ Level Structure & Tile Types

- **Grid size:** 24×24 default, expandable up to 48×48 by spending Nutrient.
- **Biomass budget** determines how many tiles can be placed.
- **Start Tile:** Bottom center of grid.
- **Goal Tile:** Heartroot; must be reachable through a valid path.
- **Validation:** System ensures the level can be solved by the creator before it is published.



Structural Tiles

- *Flesh Tile:* Basic walkable path.
- *Digest Wall:* Destructible with Acid Sac.
- *Pulse Gate:* Requires corresponding upgrade to pass.
- *Heartroot:* Goal tile, required in all levels.



Hazard Tiles

- *Acid Pool:* Deals damage to the Crawler.
- *Sticky Web:* Reduces speed unless negated.
- *Quivering Spores:* Releases damage/debuff on proximity.



Environmental Surfaces (Tile Tags)

- *Slime:* Reduces traction.
- *Water:* Slows movement or disables upgrades.
- *Soil:* Enables growth or spore-based tiles.
- *Mycelium:* May spread or mutate surrounding tiles.



Projectile Tiles

- *Guard Nodule:* Fires in a set direction at intervals.
- *Spine Shooter:* Fires in straight lines.

- *Tendrill Lash*: Hits adjacent tiles or arcs periodically.

Functional Tiles

- *Trigger Node*: Activates linked doors/traps.
 - *Membrane Tile*: Obscures content underneath.
 - *Camouflage*: Masks the tile's true type.
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Runtime Mechanics

- Game is **turn-based**, 1 turn = 1 player action.
 - Player moves 1 tile.
 - AOE hazards apply effects.
 - Trigger Nodes activate.
 - Projectiles fire or move.
 - Enemy logic (future update) resolves.
 - Deterministic flow aids in debugging and level solving.
 - Movement order creates tactical opportunity and puzzle depth.
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User Interface

Explorer Mode

- **Top Bar**:
 - Nutrient and Sporeprint counters.
 - Toggle buttons for active upgrades (1–3).
- **Bottom Corners**:
 - Restart and Menu buttons.
- **Tooltip system** for upgrades and hazards.
- Responsive design: minimalist, with animated transitions.

Builder Mode

- **Top Bar**:
 - Biomass counter with visual indicator.
- **Left Sidebar**:
 - Scrollable tile palette with icons and biomass cost.
- **Bottom Bar**:
 - Validate, Save, Publish buttons.
- Undo button in bottom-left corner.
- Tile placement via drag-and-drop or click-to-place.

- Illegal placement highlights and validation prompts.
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MVP Scope

Phase 1 Goals: - Crawler movement, tile logic, and hazard interaction. - 3 core upgrades implemented. - Basic tile palette and builder interface. - Level validation system. - JSON-based local save/load.

Excludes: - Animations beyond squash/stretch. - Online backend or profile tracking. - Cosmetic tile themes.

Asynchronous System (Post-MVP Plan)

- **Backend features:**
 - Player accounts (UUID-based).
 - Shared Coil browser with metadata: plays, likes, difficulty.
 - Automatic reward distribution.
 - Abuse prevention (IP/session checks).
 - **Community layer:**
 - Rating, comments (optional), featured levels.
 - Seasonal challenges and curated highlights.
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Visual & UX Design

- **Theme:** Alien biotech (deep teals, glowing greens, muted purples).
- **Fonts:** Rounded sans serif (e.g. Kenney Future).
- **Components:**
 - Pill-shaped buttons, translucent overlays.
 - Hover animations, squash/stretch click effects.
 - Modular UI scenes in Godot (e.g. ResourceBar, TilePalettePanel).

Post-MVP Effects: - Shader overlays for active upgrades. - Spore particles in UI. - Heartroot pulsing animation.

Design Pillars

- **Asynchronous Creative Ecosystem:** Players influence each other through level design.
- **Biotech World-Building:** A world that feels alive, strange, and reactive.
- **Progress Through Participation:** Playing and being played are both meaningful.
- **Tactical Depth + Puzzle Solving:** Each Coil is a tactical puzzle with deterministic logic.
- **Accessible Complexity:** Easy to start, rich to master, suited to both casual play and creative expression.